

Chapter 2

Fundamental Limits in Confocal Microscopy

JAMES PAWLEY

Integrated Microscopy Resource, 1675 Observatory Dr., Madison, WI 53706

INTRODUCTION

The previous chapter described how the confocal approach developed from conventional light microscopy and outlined the basic advantages to be gained by the use of confocal sampling techniques. In this chapter I will discuss the fundamental considerations that limit the performance of all confocal microscopes. Though at present no commercially available equipment approaches these limits, I will describe some simple tests to help the user assess how closely a given instrument approaches the ideal.

What Limits?

The task of the confocal light microscope is to measure the optical or fluorescent properties within a number of small, contiguous sub-volumes of the specimen (Fig. 1). The fundamental limits on this process therefore are related to the quantitative accuracy with which these measurements can be made (n_2/n_1), the size (δx , δy , δz) and position (x , y , z) of the sub-volume to be sampled, limitations imposed on the rate at which these measurements can be made by the effects of photo-damage to the specimen, source brightness and fluorescence saturation, and limitations imposed by the fact that the continuous specimen must be measured in terms of discrete volume elements called voxels (a voxel is the 3D equivalent of a pixel, which is the smallest element of a 2D image). This chapter will try to define the factors that ultimately limit the accuracy with which these measurements can be made. As such, it will serve as an introduction to a number of the chapters that follow, in which the practical and theoretical aspects of these problems will be discussed in greater detail. The discussion should be applicable to a consideration of all types of confocal microscopes, though here, as elsewhere in this volume, microscopes in which scanning is accomplished by moving the beams(s) rather than specimen will be emphasized because they are more easily applied to living specimens. In some cases, differences between the mirror-scanning and disk-scanning instruments (including both tandem and single-sided disks) will dictate a separate consideration.

The data recorded from a confocal microscope will, in the simplest case, be a set of intensity values for every voxel throughout a 3-dimensional volume within the specimen. Though the data may often be displayed as an image, it should always be remembered that the confocal microscope is intrinsically a serial or sampling instrument not a parallel-imaging instrument. While it is true that we may choose to sample a plane by sequentially scanning the illumination over all of the points in that plane and that by doing so, we may produce an image, given sufficiently flexible equipment, we could also use

the same total sampling (imaging) time to measure either a single point at a great many different times, a small volume within the sample, or indeed any collection of points within the specimen several times.

The distinction between sampling and imaging is, of course, not absolute; after all, most of us will view the final result as some sort of image. However, the distinction is still useful because it requires one to explicitly confront many problems that are not always so obvious when images are viewed visually or after photographic recording.

The sampling approach, which is covered in more detail in Chapter 4, allows an image to be built up from a number of individual measurements that reflect properties within specific regions of the sample. In the case that the measured properties are optical, the measurements involve counting photons and

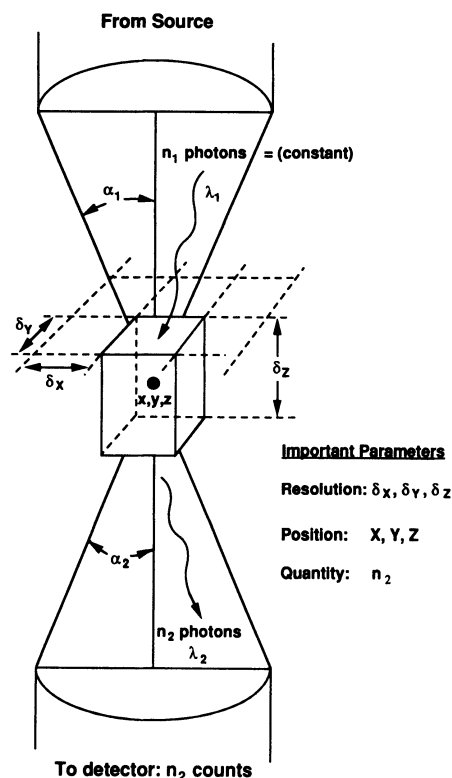


FIG. 1. Parameters related to the fundamental limitations of confocal microscopy.

this process implies limits on the accuracy and data rate usually not explicitly considered in normal microscopy. Factors that pose these limits are counting statistics, source brightness, and specimen response. These are discussed below.

Counting statistics

The accuracy of any specific measurement of fundamental quantum interactions is limited by Poisson statistics. Without going into the details, this means that if the same measurement is made repeatedly, and the average result of these measurements is N , the chance that any specific measurement is in the range of $N \pm \sqrt{N}$ is 63%. For example, if N is 100, 63% of the measurements will be in the range from $100 - \sqrt{100}$ to $100 + \sqrt{100}$ or between 90–110. Such a measurement is said to have 10% statistics. It can be seen that increasing the precision to 1% requires that N be increased to 100^2 or 10,000. While similar considerations limit the performance of all types of microscopical measurements, they are more explicit in their effect on confocal microscopy where photons are routinely counted individually.

Source brightness

There is a fundamental law of optics which states that the brightness of light in the image (measured in watts/cm²/steradian) can never be greater than that in the source. In the case of laser-scanning microscopes, the intrinsic brightness of the laser source is so high that this law does not present a practical limitation on performance (though degradation of the specimen may do so). However, it does pose a limitation on the performance of disk-scanning instruments, which currently use Hg arc sources and lose 99% of their intensity in passing through the disk. In the latter case, the source brightness and the optical design of the illuminating optics are crucial to being able to detect enough photons to produce a statistically well-defined fluorescence image in a reasonable amount of time. At present, narrow-band illumination emerging from the best disk-scanning instruments is at least an order of magnitude below that needed for them to compare with the laser instruments. Further improvement may require using large (5w), phase-randomized laser sources as described in Chapters 1 and 6.

Specimen response

In most cases it is safe to assume that photons interact with the specimen in a manner which is independent of the intensity of the illumination. However, this linear response may not be maintained at the high levels possible in laser-based confocal microscopes (see Chapter 16). Conceivable departures include the possibility that absorption in the specimen may cause sufficient warming to produce damage or that the electric field strength may become sufficient to produce a non-linear response, but the most obvious complication is the phenomenon of fluorescence saturation.

This phenomenon occurs when the flux of exciting illumination is so intense that, at any instant, a significant fraction of the fluorescent molecules are in the excited state. As excited molecules no longer absorb light at the usual wavelength, this has the effect of lowering the effective dye concentration and

reducing the fluorescent efficiency. This saturation threshold can easily be reached at the very high flux levels found in laser-scanning confocal microscopes (see Chapters 3, 5 and 16). The problem is more severe when using dye molecules with long fluorescent decay times or when the dye must be used at low concentration.

This fundamental limitation on the rate of fluorescent data acquisition can be side-stepped if the illuminating light is formed into more than one focussed spot on the specimen. However, to preserve confocal conditions, a separate pinhole and detector must be used for each spot, a condition present in the disk-scanning instruments. In addition to saturation limitations on data transfer rates, high light fluxes may effect the rate at which the fluorescent dye is bleached through a variety of complex mechanisms that while as yet imperfectly understood may in the end be sensitive to dose rate. (see Chapter 16).

A Typical Problem

To highlight the interactions between the limits imposed by counting statistics and more widely recognized limits such as spatial resolution, let us define a characteristic microscopical problem shown schematically in Fig. 2. The specimen is a cell in which some of the protein sub-units making up the cytoskeletal fibers have been replaced with fluorescent analogs. These fibers are very small, about ten times smaller than the resolution limit of light optics, and excitation of the fluorescent dye causes it to bleach, a process that is probably toxic to the cell. Because the object is to observe the formation and movement of the linear cytoskeletal elements within the living cell, we cannot take advantage of the higher spatial resolution of the electron microscope.

This example, though perhaps overly specific, is not uncharacteristic, and it has the advantage of highlighting most of the fundamental limitations that I wish to discuss. The important features of this example are as follows:

- 1) Observations will be improved by high spatial resolution.
- 2) Accuracy in measuring intensity will be important as it may permit determination of the number of linear polymers bundled together in each visible fiber.

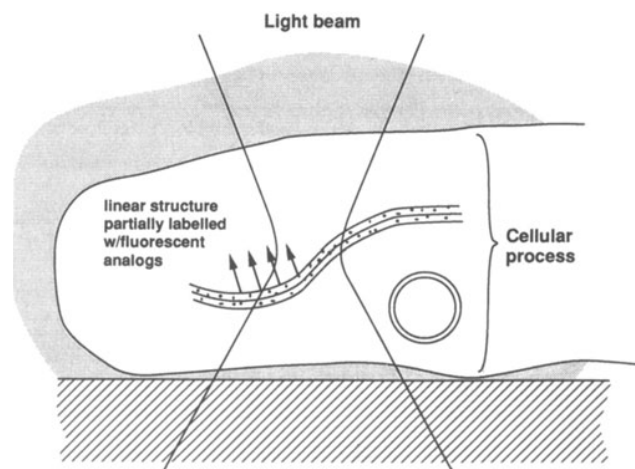


FIG. 2. Diagram of a notional specimen: a living cell process containing a filamentous structure smaller than the resolution limit and sparsely stained with fluorescent analog molecules.